

A Dataset of Synchronized Raw and Preprocessed Finger-Contact ECG and Dual-Wavelength PPG Signals from Healthy Subjects at Rest and During Seated Post-Exercise Recovery

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Abstract

Electrocardiogram (ECG) and photoplethysmogram (PPG) signals are widely used noninvasive methods for assessing cardiovascular activity and provide complementary information about the cardiac cycle. ECG records cardiac electrical activity, whereas PPG records optically detected blood-volume changes in peripheral tissue. This paper describes a synchronized ECG-PPG dataset collected from 148 apparently healthy subjects under a controlled seated protocol at rest and during post-exercise recovery after two treadmill-running conditions. Signals were acquired using a custom card-type handheld finger-contact prototype that records single-lead ECG and dual-wavelength PPG at 660 nm and 940 nm concurrently. The dataset contains 444 condition-specific records, with each subject contributing one seated resting record, one seated recovery record after light treadmill running, and one seated recovery record after moderate treadmill running. Both raw ADC-count signals and preprocessed signals are provided, and the accompanying software and example code are publicly available. The dataset is intended for research on synchronized ECG-PPG signal analysis, waveform-quality assessment, controlled post-exercise recovery physiology, and exploratory PPG-to-ECG reconstruction under controlled conditions. It should not be interpreted as a free-living wearable dataset or as clinical diagnostic ECG ground truth without external validation.

Dataset: A dataset of simultaneous collected ECG and PPG signals: <https://data.mendeley.com/datasets/pz2zzr8vhm/1> (accessed on 3 March 2026).



Academic Editor: Pufeng Du

Received: 18 April 2026

Revised: 19 June 2026

Accepted: 21 June 2026

Published: 23 June 2026

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Keywords: photoplethysmography; electrocardiogram; physical exercise; biomedical dataset

1. Summary

The World Health Statistics Report 2024 indicates that cardiovascular diseases (CVDs) remain the leading cause of death worldwide [1]. Electrocardiography (ECG), as the core method of measuring cardiac electrical activity, has become the gold standard for vital-signs monitoring and CVD diagnosis because it is simple, inexpensive and rapid [2].

However, achieving long-term continuous acquisition of high-quality ECG data in everyday settings remains a major research challenge. The growing use of wearable devices such as smartwatches has opened new avenues for cardiovascular monitoring, but these devices have clear limitations for ECG surveillance, including the need for active user cooperation and susceptibility to routine-life interference, which restrict their broad application.

A complete single cardiac cycle of an ECG signal consists of three key components: the P wave, the QRS complex, and the T wave. Figure 1 illustrates a typical ECG waveform and its major morphological features. Figure 1a presents core elements within one cardiac cycle, including the P wave, the QRS complex, and the T wave, as well as the PR interval, the QT interval, the PR segment, and the ST segment. The RR interval between two adjacent cardiac cycles is also marked in the figure. Photoplethysmography (PPG) is an optical sensing method that captures dynamic changes in blood volume within human tissues. A standard fingertip PPG waveform comprises an ascending systolic limb, systolic peak, dicrotic notch, and diastolic descending limb. These morphological features collectively reflect pulse propagation timing, vascular tone, and peripheral tissue perfusion. Red and near-infrared light are the most commonly used light sources for PPG, as they exhibit distinct optical absorption characteristics in human tissues and blood components. Figure 1b displays a PPG waveform over one cardiac cycle with four key feature points labeled A, B, C, and D. Specifically, Point A corresponds to the primary systolic wave, Point B to the tidal wave (precursor of the dicrotic wave), Point C to the peak of the dicrotic rebound wave, and Point D to the dicrotic trough. The relative amplitudes and temporal relationships among these four landmarks can reflect pulse-wave morphology and peripheral hemodynamic changes.

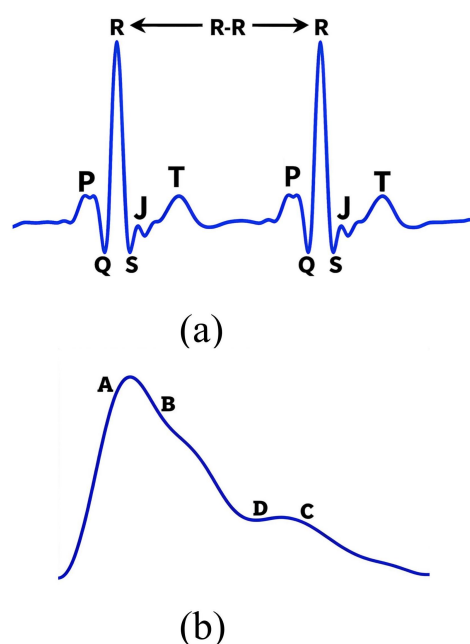


Figure 1. Typical ECG and PPG signals. (a) ECG signal and its features. (b) PPG signal and its waveform features.

ECG and PPG signals share periodicity from a physiological perspective: ECG reflects the heart's electrophysiological activities, while PPG characterizes the mechanical pulse transmitted to peripheral tissues via blood flow. Some timing-related ECG features are associated with PPG pulse characteristics. PPG-to-ECG reconstruction is framed as an exploratory algorithmic research task under controlled acquisition conditions. Such work may support investigation of ECG-PPG associations, but it does not establish continuous clinical monitoring, early CVD screening, or intervention capability. Banerjee et al. explored

the correlations between segmented ECG features and PPG characteristics [3,4]. Their results demonstrated that ECG-derived timing metrics including the PR interval, QRS duration, QT interval, and RR interval can be estimated from features extracted from PPG signals.

Compared with ECG, PPG features superior convenience and lower cost, making it widely integrated into consumer wearables such as smartwatches and smart rings. These devices typically collect signals from flexible sampling sites including earlobes, fingertips, and wrists [5]. Driven by the rapid development of artificial intelligence, numerous studies have focused on PPG-to-ECG reconstruction using diverse algorithms. Some researchers extract time- and frequency-domain features from PPG signals and establish mapping relationships between these features and ECG parameters (PR interval, QRS interval, QT interval, RR interval), as well as features in the discrete cosine transform (DCT) domain [3,6]. Other studies adopt deep learning to achieve direct end-to-end ECG reconstruction from raw PPG data [7–12]. For instance, Nirmal et al. [13] proposed a deep learning-based PPG-to-ECG reconstruction model (P2ERM) and deployed it on an ARM v6 controller. In our previous work, we verified the feasibility of deep learning-based ECG reconstruction from PPG using subsets of the MIMIC-III and MIMIC-II databases, and achieved promising results [14–18]. Figures S1 and S2 present the flowchart and the proposed algorithm model constructed in our previous study for ECG signal reconstruction from PPG signals.

Although PPG-to-ECG reconstruction is an active exploratory research topic for wearable-oriented algorithms, its practical performance and robustness under real wearable scenarios have not been fully validated. The lack of dedicated public datasets has become a major bottleneck in this research field. As summarized in Table 1, mainstream public ECG and PPG datasets such as the MIMIC series and VitalDB are mainly collected via bedside monitors in intensive care units and operating rooms, with data primarily obtained from critically ill bedridden patients. These datasets only record signals under resting or low-activity states and contain no data acquired during exercise. Such differences in acquisition environments, participant cohorts, and signal characteristics lead to a large domain gap between these public datasets and real-world wearable data.

Table 1. Comparison of public ECG-PPG or adjacent physiological datasets.

Dataset	Acquisition Setting	Subjects/Records	Modalities	Relevance and Limitation
MIMIC II [19]	Hospital ICU, resting/bedridden	Large ICU population	ECG, PPG, etc.	Large clinical resource; not wearable acquisition.
MIMIC-III subset [20]	Hospital ICU, resting/bedridden	10,282 patients	ECG, PPG, ABP	Useful clinical waveform resource.
VitalDB [21]	Operating room/critical care	6090 patients	ECG, PPG, ABP, etc.	High-fidelity surgical database; not for wearable use.
Wrist PPG [22]	Active exercise, separate devices	8 subjects, 19 records	ECG, PPG, ACC	Motion data available; needs alignment, small sample.
Sens Smart Tech [23]	Synchronous post-exercise test	32 subjects, 338 records	ECG, PCG, PPG, SCG	Post-exercise dataset; large equipment.
Our dataset	Seated rest and post-exercise recovery, finger prototype	148 subjects, 444 records	Single-lead ECG, PPG	Synchronized raw and preprocessed ECG-PPG from one device.

Existing public datasets containing exercise-related data also have obvious drawbacks. The Wrist PPG during Exercise dataset released by Jarchi et al. [22] contains wrist-worn PPG and chest-lead ECG recordings collected during walking, running, moderate cycling, and high-intensity cycling. Nevertheless, ECG and PPG signals were captured by separate devices, resulting in inherent synchronization errors. Signal alignment had to be indirectly implemented based on timestamps. Additionally, this dataset has a small sample size, and the recorded signals suffer from severe noise during exercise. Another dataset named Sens Smart Tech, published by Lazoviki et al. [23], contains synchronously acquired ECG, phonocardiogram, PPG and cardioseismograph signals after exercise. However, its bulky and complex acquisition system is inconsistent with the form factor of consumer-grade wearables, limiting its applicability for wearable algorithm research.

In this work, we construct and release a publicly available dataset of synchronously recorded ECG and PPG signals under three physiological conditions: resting state, post-light-exercise recovery, and post-moderate-exercise recovery. Using a custom card-type handheld acquisition system, we synchronously collect single-lead finger ECG and finger PPG signals from all participants under these three conditions. The proposed dataset provides a controlled finger-contact benchmark for physiological signal analysis and exploratory PPG-based ECG reconstruction; it is not intended to demonstrate performance in ambulatory consumer wearables. Compared with existing public ECG–PPG datasets, the present work has four main advantages. First, the dataset focuses on synchronized ECG–PPG acquisition under post-exercise recovery conditions, helping to fill a public-data gap for controlled exercise-related ECG–PPG algorithm research. Second, it includes 148 valid participants, each with paired ECG and PPG signals recorded under three physiological conditions. This within-subject repeated-condition design supports systematic investigation of exercise-related changes in physiological signal characteristics. Third, the hardware design enables simultaneous finger-contact ECG and dual-wavelength PPG acquisition in a compact handheld device. The recorded signals retain contact-related noise characteristics that are useful for algorithm robustness testing under this specific protocol. Fourth, by targeting healthy participants under exercise-related scenarios, this dataset complements classical public datasets such as the MIMIC series and VitalDB in terms of acquisition setting, participant population, physiological state. These complementary characteristics provide a useful controlled resource for exploratory PPG-to-ECG reconstruction and related physiological signal analysis.

The rest of this paper is organized as follows. Section 2 details the overall structure and content of the proposed dataset. Section 3 fully introduces the data acquisition hardware, experimental workflow, offline signal processing pipelines and signal quality evaluation criteria. Section 4 presents dataset structural features, signal quality metrics, and valid application scope. Section 5 systematically summarizes dataset usage guidelines, applicable scenarios, and inherent limitations. Finally, Section 6 concludes this study, clarifies research boundaries, and puts forward prospects for future algorithm development and external validation.

2. Data Description

The dataset described in this paper is available in the Mendeley Data repository and occupies approximately 616 M Bytes of storage space. In the process of organizing the dataset, we removed the personal information of each subject such as name, contact information, etc., to protect their privacy and named the data by ID plus number such as ID1, ID2, ..., ID148. The term valid record in this paper denotes a condition-specific CSV record that contains parseable ECG, red-light PPG, and infrared-light PPG channels with sampling-rate metadata and was retained after anonymization and dataset organization.

Signal validity in this sense is different from high signal quality; users should still apply task-specific quality screening.

Data were collected from the School of Life and Environmental Sciences, Guilin University of Electronic Technology, Guilin City, China. The total dataset includes 148 apparently healthy subjects by self-report and 444 condition-specific physiological records, corresponding to 3 records per subject. The three acquisition conditions are seated rest, seated recovery immediately after light treadmill running, and seated recovery immediately after moderate treadmill running. ECG and PPG were not recorded during active treadmill movement. The collection period was from 11 October to 31 December 2025. Each record in this dataset consisted of 2 wavelengths of red ($\lambda = 660$ nm) and infrared ($\lambda = 940$ nm) PPG signals and ECG signals with 1-lead. Blood pressure was measured using a Myriad EMP10 monitor only in the resting state. Therefore, SBP and DBP are resting-state metadata and should not be used as post-exercise blood-pressure labels. Participants' demographic data were also collected in this process, and all data files were stored in comma-separated values (CSVs) format. The gender variable indicates the sex of each subject and the type was categorical. The age of the subjects was expressed in years as a numeric data type in integer format and blood pressure values were expressed in mmHg as a numeric data type in integer format. Table 2 summarizes the main acquisition parameters and protocol settings used in this study, including sensor characteristics, sampling conditions, and reference measurements.

Table 2. Summary of the technical and procedural parameters used during data acquisition.

Parameter	Value/Description	Quantization/Data Type
Number of subjects	148, ID1–ID148	Integer subject identifier after anonymization
Number of condition-specific records	3 conditions per subject; 444 retained records	CSV files in <i>Raw_Data</i> and <i>Filtered_Data</i>
Sampling frequency	250 Hz	Stored as metadata in each CSV
Nominal recording duration	180 s per condition-specific record, nominal 45,000 samples per record	Raw files may be longer or shorter than 180 s; filtered files may be shorter after processing
Actual raw-record samples	max: 62,850 min: 43,475	Integer sample count
Actual filtered-record samples	max: 45,000 min: 43,475	Integer sample count
PPG wavelengths	Red 660 nm and infrared 940 nm	Raw ADC counts; filtered floating-point values
ECG acquisition	Single-lead finger-contact ECG	24-bit ADC counts in <i>Raw_Data</i> ; converted mV in <i>Filtered_Data</i>
ECG voltage calibration	Sine-wave fitting from 0.1 mV to 3 mV simulator output, 29 amplitude levels	Equation: $\text{ecg_mv} = (\text{ecg_data} - 11,872) / (2^{24}) \times 1000 \times 0.7867 + 0.0093$, $R^2 = 1.0$ and $\text{RMSE} = 0.0025$.
Acquisition site	Left index finger for PPG and ECG contact, both thumbs on top electrodes	Not passive wrist/ring acquisition
Recording conditions	Seated rest; seated recovery after 3 min at 4 km/h (0 incline); seated recovery after 3 min at 6 km/h (10 incline)	Condition code $x = 1, 2$, or 3
Reference measurements	Systolic blood pressure, diastolic blood pressure, heart rate, only at rest	Integer metadata fields in <i>information.csv</i>

Dataset Composition

As shown in Figure 2, the dataset contains several files and folders. A file named “*information.csv*” records basic information about all subjects, including unique identification

ID (e.g., 1, 2, ...148), age, gender, height, weight, systolic blood pressure, diastolic blood pressure, and heart rate. Table 3 presents the CSV file format and data dictionary.

Table 3. CSV file format and data dictionary.

File	Column	Meaning
information.csv	ID	Subject identifier ranging from 001 to 148.
information.csv	Age, Gender, Height/cm, Weight/kg	Basic demographic and anthropometric metadata.
information.csv	SBP/mmHg, DBP/mmHg, HR/bpm	Resting cuff blood pressure and resting heart rate.
Raw_data/000_x.csv	ECG_I	Raw single-lead ECG signal.
Raw_data/000_x.csv	PPG_RED	Raw red-light PPG signal.
Raw_data/000_x.csv	PPG_IR	Raw infrared-light PPG signal.
Raw_data/000_x.csv	{‘Hz’: 250}	Sampling rate marker, not a signal channel.
Filtered_Data/000_x.csv	ECG_I_Filtered	Filtered ECG signal.
Filtered_Data/000_x.csv	ECG_I_mv	ECG amplitude in millivolts after calibration conversion.
Filtered_Data/000_x.csv	PPG_RED_Filtered	Filtered red-light PPG signal.
Filtered_Data/000_x.csv	PPG_IR_Filtered	Filtered infrared-light PPG signal.
Filtered_Data/000_x.csv	Formula/Hz	Sampling rate and ECG voltage conversion formula.

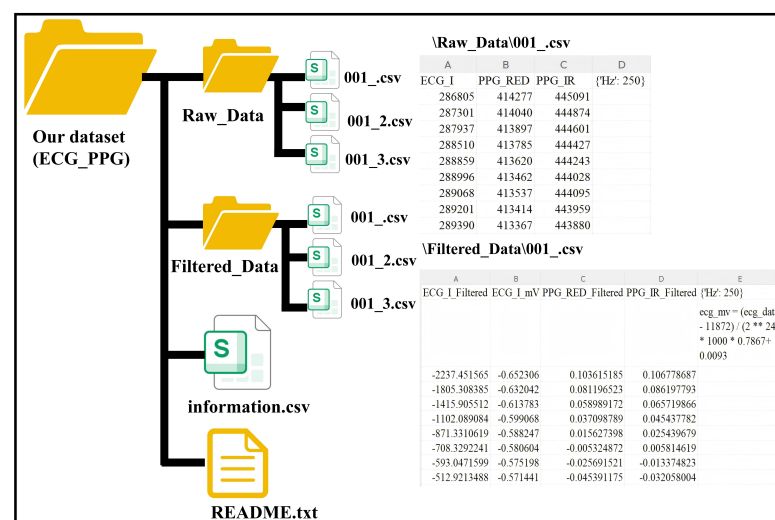


Figure 2. The electronic repository structure.

The dataset consists of two folders. The Raw_Data folder contains 444 raw waveform data files corresponding to 3 acquisition conditions for 148 subjects, in CSV format. Each file includes four signal segments. The first column, labeled ECG_I, stores the original ECG signal. The second column, labeled PPG_RED, stores the original PPG signal measured in red light. The third column, labeled PPG_IR, stores the original PPG signal measured in infrared light. The numerical data in the first three columns are of integer type and represent the original analog-to-digital converter (ADC) values. The fourth column is a string labeled {‘Hz’: 250}; it does not contain signal data and only records the sampling rate.

The Filtered_Data folder contains 444 preprocessed waveform CSV files exported by a self-developed host computer software, corresponding to three data acquisition conditions from 148 subjects. Each file contains five signal segments. Column 1, ECG_I_Filtered, stores the filtered ECG signal. Column 2, ECG_I_mv, stores the converted ECG signal to millivolts and it is floating-point type. Column 3, PPG_RED_Filtered, stores the filtered red PPG signal and is floating-point. Column 4, PPG_IR_Filtered, stores the filtered infrared

PPG signal and is floating-point. Column 5 stores the sampling rate and the ECG voltage conversion formula.

Although each experimental condition was designed for a standard 180 s recording, individual samples exhibited inconsistent durations during actual acquisition: a small number of samples were recorded for less than 180 s due to incomplete acquisition, while a few other samples exceeded 180 s in length. After data collection, invalid and low-quality signal segments were further removed during signal preprocessing, which also caused some filtered files to be shorter than the original recordings. Table 2 presents the min–max duration range and corresponding sample counts for both raw and filtered datasets.

In both the Raw_Data and Filtered_Data folders, file names follow the format 000_x, where “000” denotes the subject ID and “x” denotes the subject’s state during acquisition; x takes the values 1, 2, or 3, which correspond to three different exercise conditions. Thus, each ID has three CSV waveform files, for example, 001_1.csv, 001_2.csv and 001_3.csv, representing synchronized PPG and ECG signals recorded under three different conditions. Note that blood pressure was measured only in the resting state and not after exercise; therefore, blood pressure values appear only in the CSV file ending with “_1” for each ID. Figure 3 shows the raw ECG and dual-wavelength PPG signals and the filtered and processed signals acquired synchronously after three exercise states in one subject. If desired, users can perform the analysis using the data analysis software described in the supplementary materials below. Figure 4 shows the average heart rate values of all subjects after the three exercise states.

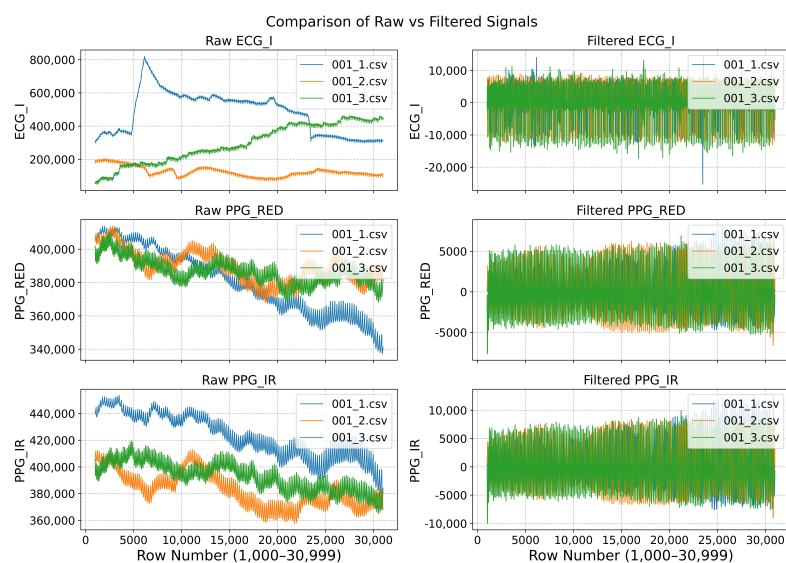


Figure 3. Raw ECG and dual-wavelength PPG signals and the filtered, post-processed signals were acquired synchronously after three exercise states for subject ID 001. The raw signals are shown on the left, and the filtered signals are shown on the right. ECG and PPG signals in the resting state are plotted as blue lines, signals after light exercise are plotted in orange, and signals after moderate exercise are plotted in green.

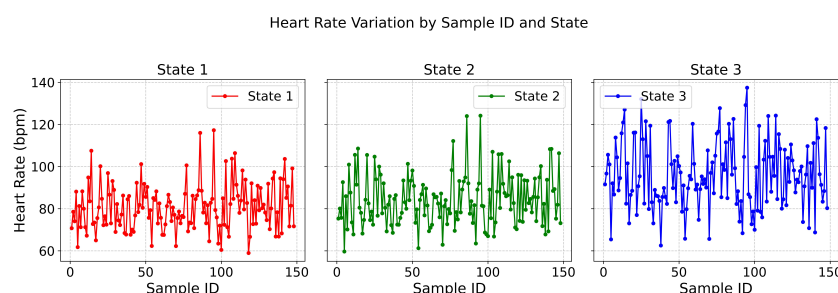


Figure 4. Mean heart rate values of all subjects after the three exercise states.

3. Methods

3.1. PPG and ECG Acquisition System

Figure 5 shows the PPG and ECG signal acquisition system. The system comprises lower-computer hardware and upper-computer software. Figure S3 illustrates the self-developed software proposed by our team. The hardware includes a dual-wavelength PPG acquisition module, a single-lead ECG module on the finger, a microcontroller unit (MCU) master control module, a Bluetooth data transmission module, and a 3D-printed enclosure; the overall dimensions are 10 cm × 6.4 cm × 1.8 cm. Three ECG electrodes are used; they are stainless steel, 3 cm × 2 cm × 0.1 cm, and accommodate the index fingers and thumbs of most users. The center of the bottom ECG electrode is hollowed out to expose the PPG sensor. Both thumbs contact the two top electrodes while the left index finger contacts the bottom electrode and covers the PPG sensor. The PPG signal is acquired in reflection mode, and the ECG is configured in a two-electrode operation according to the ECG acquisition chip manual. The accompanying data-acquisition host software was developed for PC using PyQt5.

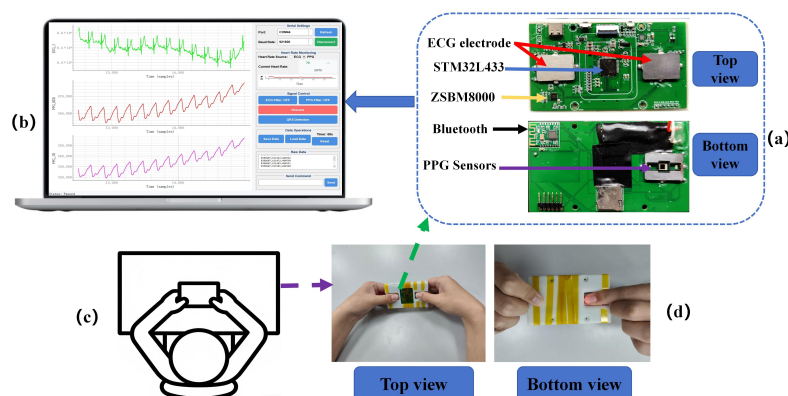


Figure 5. PPG signal and ECG signal acquisition system. (a) Device top/bottom views and hardware module layout, (b) self-developed host computer, (c) hand-contact posture, and (d) sensor contact method.

The device uses the ZSBM8000 chip (China, Zetasensing), which provides four independently configurable time slots. PPG and ECG are acquired by two separate ADCs, allowing synchronous collection of ECG signals and PPG signals. The ADC for ECG acquisition is 24-bit, offers an input impedance of 50 G Ohm, and supports a maximum differential input of ±500 mV. The chip meets the IEC 60601-2-47:2012 [24] medical standard and can be configured in a three-electrode or two-electrode mode. In three-electrode mode, the common-mode rejection ratio is 130 dB. In this device, stainless steel electrodes are used and the system is configured in two-electrode mode. The ZSBM8000 chip provides 12 LED drivers, enabling multi-wavelength PPG sensor configurations. In this device, we use light sources with wavelengths of 660 nm and 940 nm, and to improve signal quality, we drive two sets of these light sources simultaneously. The main control chip is the STM32L433 (STMicroelectronics, Geneva, Switzerland). The collected ECG and PPG data are transmitted to the upper computer software via an HC-04S Bluetooth module (Guangzhou Hui Cheng Information Technology Co., Ltd, Guangzhou, China). The system is powered by a rechargeable lithium-polymer battery of 3.7 V 500 mAh.

Synchronization and Time Alignment

ECG, red-light PPG, and infrared-light PPG are acquired within the same hardware platform and exported row-wise into the same CSV record. Therefore, within each waveform CSV file, the same sample index corresponds to the same acquisition time base for

ECG_I, PPG_RED, and PPG_IR. This design avoids the post hoc alignment required when ECG and PPG are recorded by independent devices.

Bluetooth transmission is used to transfer packaged data from the acquisition device to the host computer; it does not create separate clocks for ECG and PPG channels. Nevertheless, the present manuscript does not claim independently validated sub-sample timing accuracy. Inter-channel delay below one sample interval, absolute timestamp accuracy, long-record clock drift, and packet-loss effects should be evaluated by users whose applications require high-precision pulse transit time (PTT) estimation.

3.2. Subjects

Informed consent was obtained from all subjects. Subjects could withdraw from the study at any time. The study was approved by the Biomedical Ethics Committee of Guilin University of Electronic Technology (approval number: GUET-20251027-001). Subjects received a cash compensation of RMB 10 upon completion of the experiment. They were recruited through advertisements posted on university bulletin boards (QQ groups, WeChat groups) and through personal referrals. The subjects provided information on age, sex, height and weight; all initial data collection was carried out in a private and comfortable room. The detailed demographic and basic physiological characteristics of the participants are shown in Table 4. The public metadata file contains 148 subjects, including 85 male and 63 female subjects. Age was 23.3 ± 2.2 years (range 18.0–36.0), BMI was 21.8 ± 3.5 kg/m² (range 15.8–35.9), resting systolic blood pressure was 114.7 ± 13.6 mmHg (range 84.0–162.0), resting diastolic blood pressure was 68.0 ± 9.2 mmHg (range 52.0–95.0), and resting heart rate was 79.2 ± 9.4 bpm (range 59.0–116.0). Blood pressure and the metadata heart-rate value were measured only in the resting state. Participants were described as apparently healthy subjects based on self-report. No specialist cardiology assessment, diagnostic electrocardiography, or clinical cardiovascular screening was performed; this limitation should be considered when using the ECG channels as reconstruction targets.

Table 4. Participant metadata summary.

Variable	Summary	Notes
Number of subjects	148	All subjects have one row in information.csv.
Sex	Male 85; Female 63	Self-reported categorical metadata.
Age (years)	23.3 ± 2.2 ; range 18.0–36.0; median 24.0	Recorded as integer years.
Height (cm)	169.3 ± 8.5 ; range 150.0–188.0; median 170.0	Self-reported or measured at enrollment.
Weight (kg)	62.9 ± 13.5 ; range 40.0–110.0; median 61.0	Used to calculate BMI.
BMI (kg/m ²)	21.8 ± 3.5 ; range 15.8–35.9; median 21.3	Calculated from height and weight.
Resting SBP (mmHg)	114.7 ± 13.6 ; range 84.0–162.0; median 114.5	Measured only in the resting state.
Resting DBP (mmHg)	68.0 ± 9.2 ; range 52.0–95.0; median 67.0	Measured only in the resting state.
Resting HR (bpm)	79.2 ± 9.4 ; range 59.0–116.0; median 79.0	Recorded in information.csv.

3.3. Data Collection

In Figure 6, the data collection process for each subject is shown. Data were acquired in a temperature-controlled room to minimize perfusion changes from temperature fluctuations that could affect PPG measurements [25]. Because the device records a single-lead, card-based ECG, users must hold a designated area with both hands to capture fingertip PPG and ECG signals simultaneously. Consequently, subjects received a detailed explanation of the purpose and procedures of the experiment before wearing the device. They also learned to operate the device correctly, following the user manual, including placement of the fingers on the ECG metal electrode, contact pressure on the PPG sensor, and position of the device. Finally, each subject used the device to record a brief signal segment to confirm that the device is connected and producing the required signal, thereby ensuring

signal stability for the main experiment. After this check, each subject completed three experimental stages: Stage 1—resting state, Stage 2—motor state 1, and Stage 3—motor state 2. Subjects had to wear loose clothing and discontinue medications that might affect results, such as digitalis, before testing. They had to avoid strenuous exercise within 12 h of the session. Subjects had to provide a detailed medical history in advance, including current symptoms and any bone or joint conditions (e.g., arthritis).

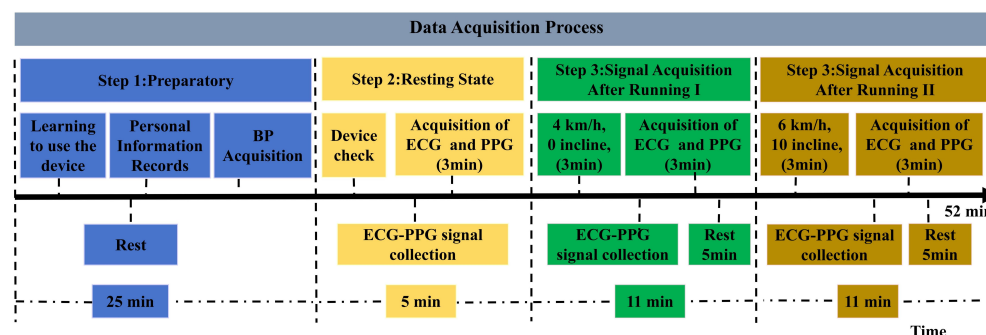


Figure 6. ECG and PPG signal data acquisition process.

Step 1: Preparatory: (1) Subjects were instructed to sit quietly and rest for 10 min and to clean their fingers to ensure signal quality; (2) subjects were required to fast for at least 2 to 3 h, avoid exercise immediately after meals, and refrain from smoking, consuming alcohol, or drinking coffee or black tea; (3) subjects were required to learn to use the data acquisition equipment for this study under the guidance of the researchers. The thumbs of both hands contacted the top 2 electrode pads, and the index finger of the left hand contacted the bottom electrode pads and made the index finger fingertips cover the PPG sensor; (4) depending on the method of using the device, each subject used the device several times for data acquisition test experiments to find the appropriate grip contact pressure of the device.

Step 2: Resting State: (1) By following the device instructions, the detector was held with both hands, maintaining a stable posture, and avoiding speaking, body movement, shaking the arm, or bending the fingers. The back was kept straight, feet flat, and arms level with the heart. (2) Distance was maintained greater than 3 m from mobile phones and other electronic devices to prevent interference. (3) Simultaneously, the ECG and PPG signals of each subject were recorded for 3 min at a sampling frequency of 250 Hz. (4) At the end of the measurement, cuff blood pressure was obtained using a Myriad EMP10 monitor.

Step 3: Signal Acquisition After Running I (Post-light recovery): Running was performed on a treadmill at a speed of 4 km/hour with an incline of 0 and a duration of 3 min. (1) Immediately after exercise, the subject held the detection device with both hands, following the data acquisition equipment instructions. The subject maintained a stable posture and did not speak, move, shake the arms, or bend the fingers. The subject kept the back straight, feet flat, and arms level with the heart. (2) During measurement, the subject remained at least 3 m away from electronic devices such as mobile phones to avoid signal interference. (3) Simultaneously, the ECG and PPG signals of each subject were recorded for 3 min at a sampling frequency of 250 Hz.

Step 4: Signal Acquisition After Running II (Post-moderate recovery): Running was performed on a treadmill at a speed of 6 km/hour with an incline of 10 and a duration of 3 min. (1) Immediately after exercise, the detection device was held with both hands, following the data acquisition device instructions, and a stable posture was maintained. The subjects did not speak, move, shake arms, or bend fingers. The back was kept straight, feet flat, and arms level with heart. (2) At least 3 m distance was maintained from mobile phones and other electronic devices during measurement to avoid signal interference.

(3) Simultaneously, the ECG and PPG signals of each subject were recorded for 3 min at a sampling frequency of 250 Hz.

All subjects were subjected to a standardized data-collection protocol consisting of the following steps: (i) a detailed explanation of informed consent was provided, (ii) written consent was obtained, (iii) demographic data were recorded, (iv) noninvasive blood pressure was measured, and (v) PPG and ECG signals were acquired across different exercise states using a homemade device. Because resting heart rates in healthy adults range from 60 to 100 bpm, the exercise protocol was designed to elicit a wider spectrum of heart rates.

3.4. Signal Preprocessing

In this study, we developed graphical user interface (GUI) data analysis software based on PyQt5, as shown in Figure S4. The software mainly includes the functions of data browsing, data interception, signal processing, data visualization, data saving, etc. Through the software, each record can be visualized and the data processing operation can be carried out through the relevant buttons. The corresponding signal processing functions of the NeuroKit2 tools (version: 0.2.12) were used in the software to process PPG and ECG data [26].

PPG bandpass filtering: PPG signals acquired by wearable devices are often susceptible to interference by motion artifacts [27], which distort the waveform. To obtain high-quality PPG signals, the sensor must maintain good contact with the skin [28]; the stability and tightness of that contact directly affect the integrity of the signal. The contact pressure exerted by the sensor on the skin can also substantially alter the shape and amplitude of the signal [29], and pressure-related discomfort may induce further distortion of the waveform. In addition, differences in local blood perfusion from the skin between individuals affect the signal-to-noise ratio and signal stability [30]. These multiple noise sources can seriously impede accurate identification of key PPG characteristic points (for example, peaks, valleys, and inflection points) and therefore reduce the reliability of subsequent physiological parameter estimates, such as blood pressure. To suppress low-frequency baseline drift and high-frequency noise and thus improve signal quality, the software includes four filter types (butter, cheby1, cheby2, and ellip), with parameters configurable via the “PPG: Params” button. Yongbo Liang et al. [31] showed that the Chebyshev II filter most effectively improves the quality of the PPG signal and identified an optimal order of 4; therefore, this study used a fourth-order Chebyshev II band-pass filter with a passband of 0.2 Hz to 10 Hz.

ECG bandpass filtering: In this study, the sources of noise contamination in the ECG data were power-line interference, electrode contact noise, motion artifacts, muscle contraction, baseline wandering, and random noise. To reduce these interferences, we used the `ecg_clean` function from the NeuroKit2 library. The `ecg_clean` function provides six ECG filtering methods: “neurokit”, “biosppy”, “pantompkins1985”, “hamilton2002”, “elgendi2010”, “engzeemod2012” and “vg”, which users can select via the ECG: Params button in the software. For this study, we used the “neurokit” option, which applied a Butterworth filter with 0.5 Hz high-pass (order = 5) followed by power-line filtering.

Raw ECG signals are converted to voltage signals: The ECG signals acquired in this study were recorded as 24-bit ADC values and therefore required conversion to voltage units. Using the acquisition device in conjunction with an ECG simulator and software (SECG1203, Beijing YeBei Technology Co., Ltd, Beijing, China), sinusoidal signals spanning 29 amplitude ranges (0.1–3 mV) were generated, and data fitting was performed to derive the conversion formula from ADC counts to voltage. The resulting fit is:

$$\text{ecg_mv} = (\text{ecg_data} - 11872) / (2^{24}) \times 1000 \times 0.7867 + 0.0093,$$

where `ecg_data` denotes the raw ECG ADC value and 11,872 represents the DC offset. The fit yielded $R^2 = 1.0$ and $RMSE = 0.0025$. Figure S5 provides an example displaying synchronously collected PPG and ECG signals as well as ECG signals converted to voltage amplitudes via our self-developed data analysis software.

3.5. PPG and ECG Signal Quality Assessment

PPG and ECG are used to estimate many physiological parameters, including heart rate, respiratory rate, and blood pressure. The accuracy of these estimates depends on the quality of the signal. When signal quality is poor, the resulting estimates are meaningless. In wearable devices, PPG and ECG signals are degraded by various interferences that produce poor signal quality. In this study, signal quality is evaluated using the functions `ecg_quality()` and `ppg_process()` of the NeuroKit2 tools. The quality of the ECG is assessed using the “averageQRS” method, which scales the quality score from 0 (lowest) to 1 (highest). The quality of the PPG was evaluated using the “templatematch” method, which computes a continuous quality index by calculating the correlation coefficient between each pulse wave and an average (template) pulse wave shape. The index is relative: 1 denotes the pulse waves most similar to the template and 0 denotes no correlation with the template. As shown in Figure 7, the ECG signal quality score was 0.866 ± 0.086 , the red-light PPG signal quality score was 0.963 ± 0.047 , and the infrared-light PPG signal quality score was 0.974 ± 0.031 .

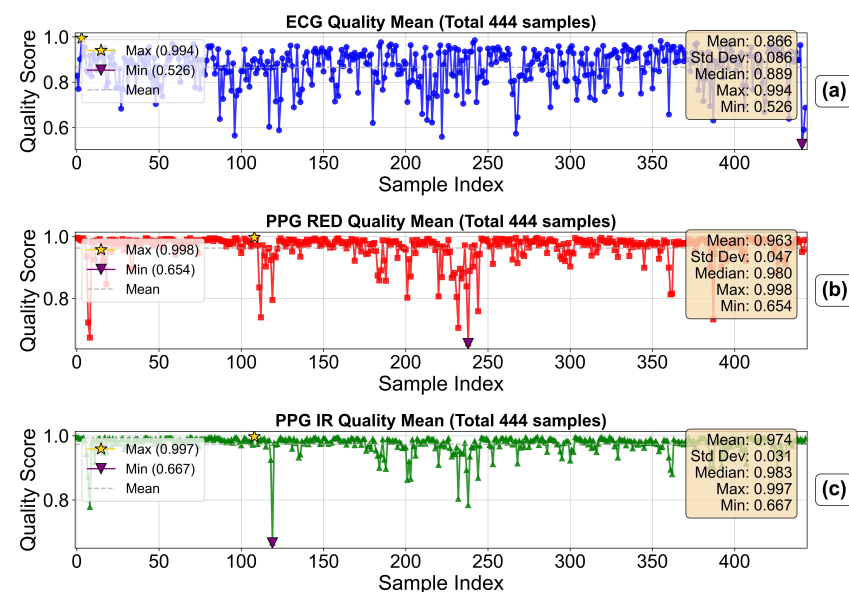


Figure 7. The results of the quality assessment for all ECG and PPG signals. (a) ECG quality, (b) red-light PPG quality, and (c) infrared-light PPG quality.

4. Descriptive Results and Discussion

The dataset provides a complete within-subject repeated-condition structure: each subject has one resting record, one post-light-exercise recovery record, and one post-moderate-exercise recovery record. This pairing facilitates within-subject comparisons of waveform quality and heart-rate-related changes after exercise. The dataset’s primary value lies in the simultaneous ECG–PPG acquisition under a controlled finger-contact protocol, not in free-living monitoring or clinical diagnosis.

Signal quality was assessed using NeuroKit2 functions. ECG quality was evaluated with the “averageQRS” method, which returns a score from 0 to 1 based on similarity to an average QRS template. PPG quality was evaluated with the `templatematch` method, which compares individual pulse waves with an average pulse-wave template. As shown

in Figure 7, PPG quality scores are high on average in this controlled seated finger-contact protocol. ECG quality was lower and more variable, which is consistent with the use of stainless steel finger electrodes and two-electrode operation. These quality scores supported dataset screening but did not convert the single-lead finger-contact ECG into a clinical-grade ground truth.

Appropriate applications include ECG-PPG synchronization studies under controlled finger-contact conditions, PPG signal-quality analysis, exploratory PPG-to-ECG waveform modeling with clear non-clinical framing, robustness testing across rest and post-exercise recovery, and educational demonstrations of multimodal physiological acquisition. Cuffless blood pressure research should be limited to exploratory resting-state analyses because SBP and DBP were not measured after the two exercise conditions.

The dataset may support future smartwatch or smart-ring research by providing a controlled benchmark for algorithm development before free-living validation. However, no smartwatch or smart-ring recordings are included, and no preliminary results from such devices should be inferred from this dataset. Future work should test trained models on independent wrist-worn and ring-worn datasets with reference ECG and reference blood pressure when those claims are made.

5. User Notes and Limitations

Users should treat this dataset as a controlled handheld finger-contact ECG-PPG dataset rather than an ambulatory wearable dataset. Consistent with the research of Rehman et al. [32], which positioned its ERP-driven XR system as an exploratory prototype instead of commercial-ready technology, we limit our work to controlled experimental research and exploratory algorithm development. This dataset should not be used to claim performance under routine daily motion, loose contact, passive wrist sensing, or long-term free-living monitoring without external validation. The ECG channel is a single-lead finger-contact signal recorded with stainless steel electrodes and two-electrode operation. It was not validated against a simultaneous clinical-grade ECG acquisition chain. Therefore, it can serve as the simultaneously acquired target waveform for algorithm research, but it should not be treated as diagnostic ground truth for cardiac morphology, clinical intervals, arrhythmia interpretation, or cardiac-state assessment. All recordings were obtained from one sensor platform, the ZSBM8000-based custom device. This design avoids cross-device timestamp matching within the dataset but introduces device-specific bias. The dataset instead provides a single-platform synchronized benchmark that requires external-device validation for generalization studies. The dataset includes healthy subjects as defined by consent, protocol completion, and self-reported screening information. Specialist cardiac assessment was not performed. Users should not use the dataset to evaluate clinical screening, diagnosis, intervention, or disease classification unless additional clinically validated labels are added. The file format is deliberately simple. Metadata in CSV format can be read by MATLAB (R2013b and above), Python (3.6 and above), all versions of R, and various spreadsheet tools such as Excel 2016. Before analysis, users should remove unstable contact periods, evaluate signal quality, document exclusion criteria, and report the usable duration for each selected record.

6. Conclusions

This data descriptor introduces a publicly available dataset comprising 444 synchronized finger-contact ECG and dual-wavelength PPG recordings. The data were acquired from 148 healthy subjects under resting conditions and seated post-exercise recovery states. This dataset includes raw and processed data files, row-wise ECG-PPG alignment generated via a single acquisition platform, comprehensive subject metadata, and quantitative

signal quality metrics. It primarily serves as a controlled research resource for investigating ECG–PPG temporal correlation, assessing signal quality, and developing exploratory PPG-to-ECG conversion algorithms. Equally noteworthy are its inherent limitations: the recordings were not collected in free-living settings, nor obtained via commercial smart-watches or smart rings. Furthermore, the dataset is not intended for clinical diagnosis, cannot act as a benchmark for post-exercise blood pressure analysis, and lacks independent validation regarding sample-level timing error and clinical ECG voltage accuracy.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/data11070155/s1>, Figure S1: Flowchart of the reconstruction of ECG signals from PPG signals; Figure S2: The architecture of the proposed Unet-BiLSTM model; Figure S3: upper-computer software; Figure S4: Data analysis software; Figure S5: Synchronized PPG and ECG signals from the ZSBM8000 chip and ECG signals converted to voltage (after filtering).

Author Contributions: Conceptualization, S.L., J.P., Q.T. and Z.C.; Methodology, S.L. and J.P.; Software, J.P. and Q.T.; Validation, C.G., Y.G., S.L. and Q.T.; Formal analysis, C.G. and Y.G.; Investigation, S.L., C.G. and Y.G.; Resources, Z.D. and Q.T.; Data curation, S.L. and J.P.; Writing—original draft, S.L.; Writing—review and editing, Z.D., J.P., Q.T. and Z.C.; Visualization, Y.G.; Supervision, J.P., Q.T. and Z.C.; Project administration, Z.C.; Funding acquisition, Q.T. and Z.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Central Guided Local Science and Technology Development Fund Project for Science and Technology Innovation Base Construction No. Guike ZY24212046, supported by the Department of Science and Technology of Guangxi Zhuang Autonomous Region. The Project for Enhancing the Basic Research Capacity of Young and Middle-aged Teachers in Colleges and Universities in Guangxi, No. 2024KY0227, supported by the Department of Education of Guangxi Zhuang Autonomous Region. Innovation Project of GUET Graduate Education No. 2024YCX5124, supported by Guilin University of Electronic Technology.

Institutional Review Board Statement: The study was approved by the Biomedical Ethics Committee of Guilin University of Electronic Technology (approval number: GUET-20251027-001).

Data Availability Statement: A dataset of simultaneous collected ECG and PPG signals, <https://data.mendeley.com/datasets/pz2zzr8vbm/1> (accessed on 3 March 2026).

Acknowledgments: This work was supported by the Guangxi Engineering Technology Research Center for Non-invasive Detection of Human Physiological Information, the Guangxi Key Laboratory of Metabolic Reprogramming and Intelligent Medicine for Chronic Diseases, and the Key Laboratory of Biomedical Sensors and Intelligent Instruments at Guilin University of Electronic Technology.

Conflicts of Interest: The authors declare no conflicts of interest.

Code Availability Statement: The associated software can be accessed via GitHub: <https://github.com/Lishiyong0607/ECG--PPG-Software> (accessed on 12 June 2026). Additional codes are available from the corresponding author upon reasonable request.

References

1. World Health Organization. *World Health Statistics 2024: Monitoring Health for the SDGs, Sustainable Development Goals*; World Health Organization: Geneva, Switzerland, 2024.
2. Le, T.; Clark, I.; Fortunato, J.; Sharma, M.; Xu, X.; Hsiai, T.K.; Cao, H. Electrocardiogram: Acquisition and Analysis for Biological Investigations and Health Monitoring. In *Interfacing Bioelectronics and Biomedical Sensing*; Cao, H., Coleman, T., Hsiai, T.K., Khademhosseini, A., Eds.; Springer International Publishing: Cham, Switzerland, 2020; pp. 117–142. [\[CrossRef\]](#)
3. Banerjee, R.; Sinha, A.; Choudhury, A.D.; Visvanathan, A. PhotoECG: Photoplethysmography to estimate ECG parameters. In *Proceedings of the 2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*; IEEE: New York, NY, USA, 2014.
4. Banerjee, R.; Sinha, A.; Pal, A.; Kumar, A. Estimation of ECG parameters using photoplethysmography. In *Proceedings of the 13th IEEE International Conference on BioInformatics and BioEngineering*; IEEE: New York, NY, USA, 2013.

5. Tamura, T.; Maeda, Y.; Sekine, M.; Yoshida, M. Wearable photoplethysmographic sensors—Past and present. *Electronics* **2014**, *3*, 282–302. [CrossRef]
6. Zhu, Q.; Tian, X.; Wong, C.W.; Wu, M. ECG reconstruction via PPG: A pilot study. In *Proceedings of the 2019 IEEE EMBS International Conference on Biomedical & Health Informatics (BHI)*; IEEE: New York, NY, USA, 2019; pp. 1–4.
7. Seo, C.; Kwon, H.; Baek, J.; Lee, S.; Lee, J.; Lee, Y.; Joo, S.; Kim, H.K.; Lee, C. Deep Learning Based Method to Electrocardiogram Reconstruction via Photoplethysmography. *J. Electr. Eng. Technol.* **2025**, *20*, 4397–4408. [CrossRef]
8. Skoric, J.; D'Mello, Y.; Plant, D.V. Generative Reconstruction of Multimodal Cardiac Waveforms From a Single Vibrational Cardiography Sensor. *IEEE J. Biomed. Health Inform.* **2025**, *29*, 6576–6587. [CrossRef] [PubMed]
9. Ezzat, A.; Omer, O.A.; Mohamed, U.S.; Mubarak, A.S. ECG signal reconstruction from PPG using a hybrid attention-based deep learning network. *EURASIP J. Adv. Signal Process.* **2024**, *2024*, 95. [CrossRef]
10. Cerda-Dávila, D.A.; Reyes, B.A. Exploring the Reconstruction of Electrocardiograms from Photoplethysmograms via Deep Learning. In *Proceedings of the 2023 IEEE EMBS R9 Conference*; IEEE: New York, NY, USA, 2023; pp. 1–4.
11. Vo, K.; Naeini, E.K.; Naderi, A.; Jilani, D.; Rahmani, A.M.; Dutt, N.; Cao, H. P2E-WGAN: ECG waveform synthesis from PPG with conditional wasserstein generative adversarial networks. In *Proceedings of the 36th Annual ACM Symposium on Applied Computing*; Association for Computing Machinery: New York, NY, USA, 2021; pp. 1030–1036.
12. Chiu, H.Y.; Shuai, H.H.; Chao, P.C.P. Reconstructing QRS complex from PPG by transformed attentional neural networks. *IEEE Sens. J.* **2020**, *20*, 12374–12383. [CrossRef]
13. Murmu, N.; Gupta, R.; Sharma, K.D. Real-time PPG-to-ECG reconstruction model with on-device recalibration facility. *IEEE Trans. Instrum. Meas.* **2024**, *73*, 4010309. [CrossRef]
14. Tang, Q.; Chen, Z.; Guo, Y.; Liang, Y.; Ward, R.; Menon, C.; Elgendi, M. Robust reconstruction of electrocardiogram using photoplethysmography: A subject-based Model. *Front. Physiol.* **2022**, *13*, 859763. [CrossRef] [PubMed]
15. Guo, Y.; Tang, Q.; Chen, Z.; Li, S. Unet-bilstm: A deep learning method for reconstructing electrocardiography from photoplethysmography. *Electronics* **2024**, *13*, 1869.
16. Guo, Y.; Li, S.; Chen, Z.; Tang, Q. Subject-independent model for reconstructing electrocardiography signals from photoplethysmography signals. *Appl. Sci.* **2024**, *14*, 5773. [CrossRef]
17. Guo, Y.; Tang, Q.; Li, S.; Chen, Z. Reconstruction of missing electrocardiography signals from photoplethysmography data using deep neural network. *Bioengineering* **2024**, *11*, 365. [CrossRef] [PubMed]
18. Tang, Q.; Chen, Z.; Ward, R.; Menon, C.; Elgendi, M. PPG2ECGps: An end-to-end subject-specific deep neural network model for electrocardiogram reconstruction from photoplethysmography signals without pulse arrival time adjustments. *Bioengineering* **2023**, *10*, 630. [PubMed]
19. Saeed, M.; Villarroel, M.; Reisner, A.T.; Clifford, G.; Lehman, L.W.; Moody, G.; Heldt, T.; Kyaw, T.H.; Moody, B.; Mark, R.G. Multiparameter Intelligent Monitoring in Intensive Care II: A public-access intensive care unit database. *Crit. Care Med.* **2011**, *39*, 952–960. [CrossRef] [PubMed]
20. Johnson, A.E.; Pollard, T.J.; Shen, L.; Lehman, L.W.H.; Feng, M.; Ghassemi, M.; Moody, B.; Szolovits, P.; Anthony Celi, L.; Mark, R.G. MIMIC-III, a freely accessible critical care database. *Sci. Data* **2016**, *3*, 160035. [CrossRef] [PubMed]
21. Lee, H.C.; Park, Y.; Yoon, S.B.; Yang, S.M.; Park, D.; Jung, C.W. VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. *Sci. Data* **2022**, *9*, 279. [CrossRef] [PubMed]
22. Jarchi, D.; Casson, A.J. Description of a database containing wrist PPG signals recorded during physical exercise with both accelerometer and gyroscope measures of motion. *Data* **2016**, *2*, 1. [CrossRef]
23. Lazović, A.; Atanasoski, V.; Tadić, P.; Djordjević, N.; Tiosavljević, M.; Ivanović, M.D.; Hadžievski, L.; Ristić, A.; Vukčević, V.; Petrović, J. Synchronous post-exercise electrocardiogram, phonocardiogram, photoplethysmograms and seismocardiogram. *Sci. Data* **2025**, *12*, 1452. [CrossRef] [PubMed]
24. IEC 60601-2-47:2012; Medical Electrical Equipment—Part 2-47: Particular Requirements for the Basic Safety and Essential Performance of Ambulatory Electrocardiographic Systems. International Electrotechnical Commission (IEC): Geneva, Switzerland, 2012. Available online: <https://webstore.iec.ch/en/publication/2666> (accessed on 18 June 2026).
25. Scardulla, F.; Cosoli, G.; Spinsante, S.; Poli, A.; Iadarola, G.; Pernice, R.; Busacca, A.; Pasta, S.; Scalise, L.; D'Acquisto, L. Photoplethysmographic sensors, potential and limitations: Is it time for regulation? A comprehensive review. *Measurement* **2023**, *218*, 113150. [CrossRef]
26. Makowski, D.; Pham, T.; Lau, Z.J.; Brammer, J.C.; Lespinasse, F.; Pham, H.; Schölzel, C.; Chen, S.A. NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behav. Res. Methods* **2021**, *53*, 1689–1696. [CrossRef] [PubMed]
27. Park, J.; Seok, H.S.; Kim, S.S.; Shin, H. Photoplethysmogram analysis and applications: An integrative review. *Front. Physiol.* **2022**, *12*, 808451. [CrossRef] [PubMed]
28. Sun, Y.; Thakor, N. Photoplethysmography revisited: From contact to noncontact, from point to imaging. *IEEE Trans. Biomed. Eng.* **2015**, *63*, 463–477. [CrossRef] [PubMed]

29. Chandrasekhar, A.; Yavarimanesh, M.; Natarajan, K.; Hahn, J.O.; Mukkamala, R. PPG sensor contact pressure should be taken into account for cuff-less blood pressure measurement. *IEEE Trans. Biomed. Eng.* **2020**, *67*, 3134–3140. [[CrossRef](#)] [[PubMed](#)]
30. Kyriacou, P.; Powell, S.; Langford, R.; Jones, D. Investigation of oesophageal photoplethysmographic signals and blood oxygen saturation measurements in cardiothoracic surgery patients. *Physiol. Meas.* **2002**, *23*, 533–545. [[CrossRef](#)] [[PubMed](#)]
31. Liang, Y.; Elgendi, M.; Chen, Z.; Ward, R. An optimal filter for short photoplethysmogram signals. *Sci. Data* **2018**, *5*, 180076. [[CrossRef](#)] [[PubMed](#)]
32. Rehman, A.; Lee, M.; Kim, Y.; Chae, M.S.; Mun, S. Machine Learning-Driven XR Interface Using ERP Decoding. *Electronics* **2025**, *14*, 3773. [[CrossRef](#)]

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